

Homework Assignment 1

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Question 1 $Y_i = 100 + 20X + e$, $sd = 25$ Suppose an observation on Y will be made for $X = 5$. a) Can you state the exact probability that Y will fall between 195 and 205? Explain.

No, without knowing the probability distribution of the error term we can not know exact probability.

b) If e_i follows a normal distribution, can you now state the exact probability that Y will fall between 195 and 205? Explain.

Yes, knowing the error term follows a normal distribution, we also now know that Y follows a normal distribution $P(195 < Y < 205) = P\left(\frac{Y - M}{sd} \in \left(\frac{195 - 200}{5}, \frac{205 - 200}{5}\right)\right) = P(-1 < Z < 1) = 0.6826895$

```
1 pnorm(205, 200, 5) - pnorm(195, 200, 5)
```

```
[1] 0.6826895
```

There is a 68.27% that Y will fall between 195 and 205

Question 2 $GPA = 2.11405 + 0.03883(ACT)$ (a) Obtain a 99 percent confidence interval for B_1 . Interpret your confidence interval. Does it include zero? Why might the director of admissions be interested in whether the confidence interval includes zero?

Sample size = 120, 118 df $t = 2.618$ $SE(B_1) = 0.01277$ $MOE = 2.618 \times 0.01277 = 0.03343$ (0.0054, 0.0723) This interval does not include zero. Since it does not include zero we reject the null hypothesis $B_1 = 0$. This suggests that there is a statistically significant positive relationship between act scores and academic performance at the 99% level.

(b) =Test whether or not a linear association exists between student's ACT score and GPA at the end of the freshman year. Use a level of significance of .01. State the hypothesis and conclusion.

Null Hypothesis: $B_1 = 0$ there is no linear association between ACT scores and freshman GPA Alternate Hypothesis: $B_1 \neq 0$ there is a linear association between ACT score and freshman GPA $t = 3.040$ p-value = 0.003 The p - value of 0.003 makes us reject the null hypothesis. There is sufficient evidence at the 0.01 significance level that there exists a linear association between ACT score and GPA at the 0.01 sig level

(c) Obtain a 95% interval estimate of the mean freshman GPA for students whose ACT test score is 28. Interpret your confidence interval $GPA = 2.11405 + 0.03883(28)$

```
1 gpa_data <- read.table("GPA.txt", header = TRUE)
2 model <- lm(GPA ~ ACT, data = gpa_data)
3 score_to_predict <- data.frame(ACT = 28)
4 predict(model, newdata = score_to_predict, interval = "confidence", level = 0.95)
```

	fit	lwr	upr
1	3.201209	3.061384	3.341033

[3.061, 3.341] I am 95% confident that the true mean freshman GPA for all students with act score of 28 lies between this interval.

- (d) Linda Jones obtained a score of 28 on the entrance test. Predict her freshman GPA using a 95% prediction interval. Interpret your prediction interval. The fit value is 3.201

```
1 predict(model, newdata = data.frame(ACT = 28), interval = "prediction", level = 0.95)
```

	fit	lwr	upr
1	3.201209	1.959355	4.443063

I am 95% confident that the actual freshman gpa for Linda will be between 1.959 and 4.443. Larger interval because we are predicting for individual student.

- (e) Is the prediction interval in part (d) wider than the confidence interval in part (c)? Should it be? Explain

The prediction interval in part d should be wider than c and it is wider. This is because we are predicting for the individual variation of a single student. This makes the error larger and therefore gives us a higher interval.

- (f) Plot the data, estimated regression function, and 95% pointwise confidence intervals.

```
1 library(ggplot2)
2 gpa_data <- read.table("GPA.txt", header = TRUE)
3
4 #regression model
5 model <- lm(GPA ~ ACT, data = gpa_data)
6 summary(model)
```

Call:

```
lm(formula = GPA ~ ACT, data = gpa_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.74004	-0.33827	0.04062	0.44064	1.22737

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.11405	0.32089	6.588	1.3e-09 ***
ACT	0.03883	0.01277	3.040	0.00292 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6231 on 118 degrees of freedom

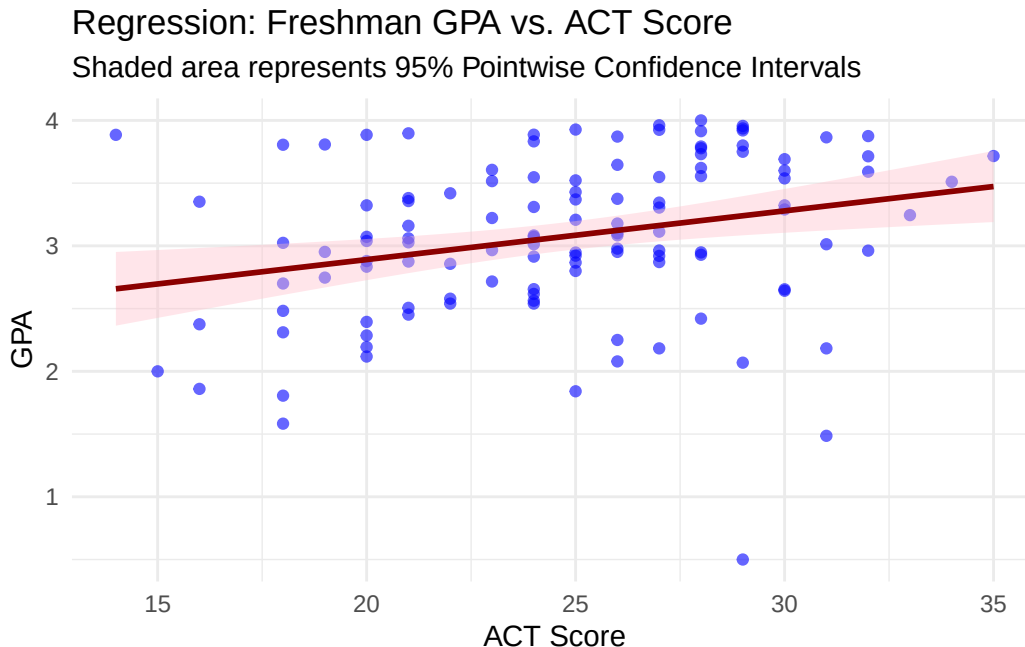
Multiple R-squared: 0.07262, Adjusted R-squared: 0.06476

F-statistic: 9.24 on 1 and 118 DF, p-value: 0.002917

```

1 ggplot(gpa_data, aes(x = ACT, y = GPA)) +
2   geom_point(color = "blue", alpha = 0.6) +
3   geom_smooth(method = "lm", level = 0.95, color = "darkred", fill = "pink") +
4   labs(title = "Regression: Freshman GPA vs. ACT Score",
5         subtitle = "Shaded area represents 95% Pointwise Confidence Intervals",
6         x = "ACT Score",
7         y = "GPA") +
8   theme_minimal()

```



g) Set up the ANOVA table.

```

1 anova(model)

```

Analysis of Variance Table

Response: GPA

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
ACT	1	3.588	3.5878	9.2402	0.002917 **
Residuals	118	45.818	0.3883		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(h) What is estimated by MSR in your ANOVA table? by MSE? Under what condition do MSR and MSE estimate the same quantity?

MSR estimates the variances of the predicted values. MSE estimates the error variance or sd^2 . They estimate the same quantity when $B_1 = 0$. This is because the slope is zero here and ACT score has no linear effect on GPA and the regression doesn't explain any extra variability.

(i) Conduct an F test of whether or not $B_1 = 0$. Set $\alpha = .01$. State the hypothesis, decision rule, and conclusion
Null Hypothesis: $B_1 = 0$ there is no linear association between ACT scores and freshman GPA
Alternate Hypothesis: $B_1 \neq 0$ there is a linear association between ACT score and freshman GPA

```
1 summary(model)
```

Call:

```
lm(formula = GPA ~ ACT, data = gpa_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.74004	-0.33827	0.04062	0.44064	1.22737

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
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ACT	0.03883	0.01277	3.040	0.00292 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6231 on 118 degrees of freedom

Multiple R-squared: 0.07262, Adjusted R-squared: 0.06476

F-statistic: 9.24 on 1 and 118 DF, p-value: 0.002917

```
1 qf(0.99, 1, 118)
```

```
[1] 6.854641
```

F-statistic: 9.24 on 1 and 118 DF, p-value: 0.00292 Since F is greater than the critical value 6.85, we reject the null hypothesis. This data tells us we are 99% certain a relationship between ACT and fresh gpa exists. However our R^2 is low meaning that act score explains about 7% of variation in GPA.

(j) Compare the F test in (i) with the t test in (b). Both tests lead me to reject the null hypothesis as both have significant p values. The $F = t^2$. F looks at variances while t looks at sd. The p-value is equal for both of these tests. One is significant if the other is.

(k) What is the absolute magnitude of the reduction in the variation of Y when X is introduced into the regression model? What is the relative reduction? What is the name of the latter measure?

The absolute magnitude is $SSR = 3.5878$ The relative reduction is SSR / SST $R^2 = 0.0726$ The name is the coefficient of determination R^2 . This value means that the act score reduces the variation in gpa by 7.26% when introduced into this model.

(l) Confirm R^2 equals the squared correlation between X and Y, and also equals the squared correlation between fitted values $\hat{Y}$ and Y

To confirm must look at R^2 because it gives us the correlation between X and Y.

```
1 cor(gpa_data$ACT, gpa_data$GPA)
```

```
[1] 0.2694818
```

```

1 fitted_values <- predict(model)
2 cor(fitted_values, gpa_data$GPA)

```

```
[1] 0.2694818
```

Both of these values squared are equal to R^2

QUESTION 3 a) Estimate 1 with a 95% confidence interval. Interpret your interval estimate.

broken = 10.0 + 4.0(transfers)

```

1 air_data <- read.table("airfreight_breakage.txt", header = TRUE)
2 colnames(air_data) <- c("broken", "transfers")
3 fit <- lm(broken ~ transfers, data = air_data)
4
5 confint(fit, "transfers", level = 0.95)

```

```

          2.5 %    97.5 %
transfers 2.942508 5.057492

```

From this confidence interval, we are 95% confident that the mean number of broken ampules lies between 2.943 and 5.057 for each additional transfer made during shipping.

- (b) Conduct a t test to decide whether or not there is a linear association between number of times a carton is transferred and number of broken ampules. Use a level of significance of .05. State the hypothesis, decision rule, and conclusion. What is the p-value of the test?

```
1 summary(fit)
```

Call:

```
lm(formula = broken ~ transfers, data = air_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2	-1	0	1	2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.0000	0.6498	15.390	1.18e-06 ***
transfers	4.0000	0.4472	8.944	4.44e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.414 on 7 degrees of freedom

Multiple R-squared: 0.9195, Adjusted R-squared: 0.908

F-statistic: 80 on 1 and 7 DF, p-value: 4.441e-05

H0: The number of times a carton is transferred has no influence on broken ampules H1: Transferring a carton does have an effect on broken ampules

At the 0.05 level, we reject the null hypothesis (p-value is 4.4e-05). This means that there is a statistically significant linear relationship between the number of transfers and the number of broken ampules.

- c) B0 represents here the mean number of ampules broken when no transfers of the shipment are made. Obtain a 95% confidence interval for B0 and interpret it.

```
1 confint(fit, level = 0.95)
```

```

                2.5 %    97.5 %
(Intercept) 8.463500 11.536500
transfers   2.942508  5.057492

```

When no transfers are made, we are 95% confident that the number of broken ampules is between 8.463 and 11.537.

- (d) (5 pt) A consultant has suggested, on the basis of previous experience, that the mean number of broken ampules should not exceed 9.0 when no transfers are made. Conduct an appropriate test, using $\alpha = 0.025$. State the hypothesis, decision rule, and conclusion. What is the p-value of the test?

H0: Number of broken ampules $B_0 \leq 9.0$ when no transfers are made HA: Number of broken ampules $B_0 > 9.0$ when no transfers are made

Reject H0 if the t^* statistic is > 2.365 or if the P value < 0.025

$$t^* = (10-9)/0.65 = 1.538$$

```
1 (1 - pt(1.538, 7))
```

```
[1] 0.08396808
```

We fail to reject H0 because the t^* value 1.538 is not greater than the critical value 2.365. and the P value 0.0839 is greater than alpha 0.025

- (e) Because of changes in airline routes, shipments may have to be transferred more frequently than in the past. Estimate the mean breakage when the numbers of transfers equal 4 with 99% confidence interval. Interpret your results.

```

1 four_transfer <- data.frame(transfers = 4)
2 predict(fit, newdata = four_transfer, interval = "confidence", level = 0.99)

```

```

      fit      lwr      upr
1  26 21.02356 30.97644

```

We can estimate that the mean number of ampules broken is 26 when 4 transfers are made

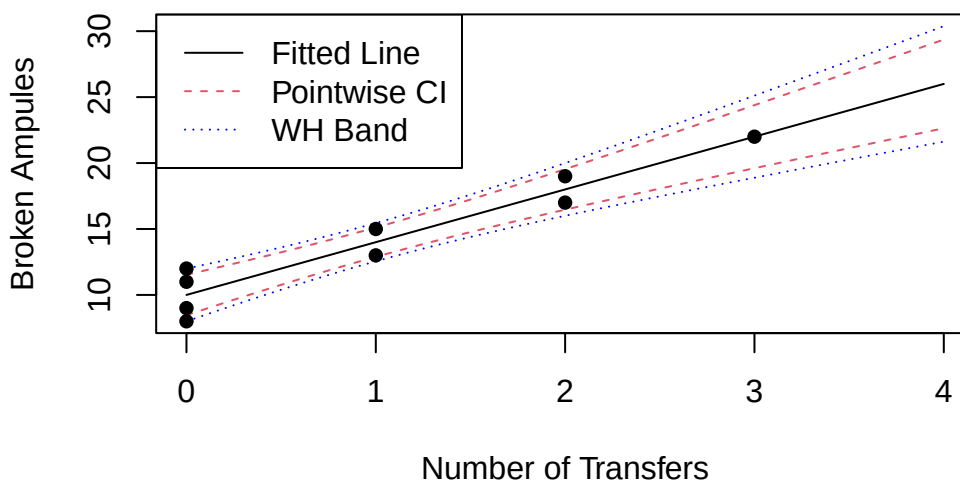
- (f) (3 pt) The next shipment will entail four transfers. Obtain a 99% percent prediction interval for the number of broken ampules for this shipment. Interpret your prediction interval.

We can be 99% confident that the mean number of broken ampules for shipments involving 4 transfers is between 21.02 and 30.98 broken ampules.

- (g) Plot the data, estimated regression function 95% pointwise confidence intervals, and 95% confidence band. Construct and plot the estimated function, confidence intervals, and confidence band in the X range from 0 to 4 (i.e. 0 to 4 transfers).

```
1 grid <- seq(0, 4, len = 100)
2 p1 <- predict(fit, newdata = data.frame(transfers = grid), interval = "confidence")
3
4 n <- nrow(air_data)
5 W <- sqrt(2 * qf(0.95, 2, n - 2))
6 # Get standard errors for the fit
7 p_se <- predict(fit, newdata = data.frame(transfers = grid), se.fit = TRUE)
8
9 band_lwr <- p_se$fit - W * p_se$se.fit
10 band_upr <- p_se$fit + W * p_se$se.fit
11
12 matplot(grid, p1, lty = c(1, 2, 2), col = c(1, 2, 2), type = "l",
13         xlab = "Number of Transfers",
14         ylab = "Broken Ampules",
15         ylim = range(p1, band_lwr, band_upr, air_data$broken),
16         main = "95% Pointwise CI vs. Confidence Band")
17
18 lines(grid, band_lwr, col = "blue", lty = 3)
19 lines(grid, band_upr, col = "blue", lty = 3)
20
21 # 6. Add original data points
22 points(air_data$transfers, air_data$broken, pch = 16)
23
24 # Legend to clarify
25 legend("topleft", legend = c("Fitted Line", "Pointwise CI", "WH Band"),
26       col = c(1, 2, "blue"), lty = c(1, 2, 3))
```

95% Pointwise CI vs. Confidence Band



Question 4 An analyst rejected conducted an F test of $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$. The p-value of the test was .033, and the analyst rejected H_0 . Was the level used by the analyst greater than or smaller than .033? If the level had been .01, what would have been the appropriate conclusion?

The alpha level would be greater than or equal to .033, so that the p - value of 0.033 was in the rejection region. If the alpha level was 0.01 there would not be evidence of a linear association between the two because $0.033 > 0.01$. The appropriate conclusion for the 0.01 level therefore there is insufficient evidence to reject the null hypothesis.